Lesson Planning

Complete Lesson Plan Book

Class: Class 12 – ICom

Subject: Business Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Functions and Limits

Learning Objectives

• Define a function and identify its domain and range.

• Evaluate limits of simple algebraic functions.

• Explain the business uses of functions.

Key Concepts

• Function: a rule pairing each input with one output.

• Domain, range and types: linear, quadratic.

• Limit: the value a function approaches.

• Cost, revenue and profit as functions of output.

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Functions and Limits to activate prior knowledge.

2. Present the core ideas of Functions and Limits with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Functions and Limits.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Functions and Limits and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Functions and Limits for classroom display.

Assessment Questions

1. Find the domain of the given function.

2. Evaluate the limit of (x^2 - 4)/(x - 2) as x approaches 2.

3. Write the profit function if R = 50x and C = 20x + 1000.

Homework

• Write a one-page summary of Functions and Limits in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)

Unit 2: Differentiation: Rules and Techniques

Learning Objectives

• Find derivatives using the power rule.

• Apply sum, product and quotient rules.

• Differentiate simple polynomial functions.

Key Concepts

• Derivative: the rate of change of a function.

• Power rule: d/dx(x^n) = n*x^(n-1).

• Sum, product and quotient rules.

• Notation: dy/dx and f'(x).

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Differentiation: Rules and Techniques to activate prior knowledge.

2. Present the core ideas of Differentiation: Rules and Techniques with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Differentiation: Rules and Techniques.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Differentiation: Rules and Techniques and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Differentiation: Rules and Techniques for classroom display.

Assessment Questions

1. Differentiate y = 3x^4 - 2x^2 + 5.

2. Find the derivative of (x^2 + 1)(x - 3).

3. Differentiate (2x + 1)/(x - 1) using the quotient rule.

Homework

• Write a one-page summary of Differentiation: Rules and Techniques in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)

Unit 3: Applications: Maxima and Minima

Learning Objectives

• Find maximum and minimum values of functions.

• Apply the second-derivative test.

• Solve profit-maximisation problems.

Key Concepts

• Critical points where f'(x) = 0.

• Second derivative test for max/min.

• Marginal cost and marginal revenue.

• Profit is maximum where MR = MC.

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Applications: Maxima and Minima to activate prior knowledge.

2. Present the core ideas of Applications: Maxima and Minima with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Applications: Maxima and Minima.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Applications: Maxima and Minima and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Applications: Maxima and Minima for classroom display.

Assessment Questions

1. Find the maximum value of y = -x^2 + 6x - 5.

2. A firm has C = x^2 + 10x + 50. Find output minimising average cost.

3. Explain the condition MR = MC for maximum profit.

Homework

• Write a one-page summary of Applications: Maxima and Minima in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)

Unit 4: Integration: Basic Rules

Learning Objectives

• Integrate simple polynomial functions.

• Apply the power rule of integration.

• Find the constant of integration using given conditions.

Key Concepts

• Integration as the reverse of differentiation.

• Power rule: integral of x^n = x^(n+1)/(n+1) + C.

• Constant of integration and initial conditions.

• Total cost from marginal cost by integration.

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Integration: Basic Rules to activate prior knowledge.

2. Present the core ideas of Integration: Basic Rules with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Integration: Basic Rules.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Integration: Basic Rules and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Integration: Basic Rules for classroom display.

Assessment Questions

1. Integrate (4x^3 + 2x - 1) dx.

2. Find y if dy/dx = 6x^2 and y = 5 when x = 1.

3. If MC = 2x + 4, find the total cost function.

Homework

• Write a one-page summary of Integration: Basic Rules in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)

Unit 5: Linear Programming

Learning Objectives

• Formulate a linear programming problem.

• Graph the feasible region of constraints.

• Find the optimal solution by the corner-point method.

Key Concepts

• Objective function to maximise or minimise.

• Constraints as linear inequalities.

• Feasible region: the polygon of allowed points.

• Optimal solution lies at a corner point.

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Linear Programming to activate prior knowledge.

2. Present the core ideas of Linear Programming with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Linear Programming.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Linear Programming and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Linear Programming for classroom display.

Assessment Questions

1. Formulate the product-mix problem from given data.

2. Graph the constraints and shade the feasible region.

3. Find maximum profit using corner points.

Homework

• Write a one-page summary of Linear Programming in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)

Unit 6: Statistics: Mean, Median and Mode

Learning Objectives

• Calculate the arithmetic mean of grouped data.

• Find the median and mode of a frequency table.

• Choose the right average for business data.

Key Concepts

• Mean: sum of values divided by their number.

• Median: the middle value; mode: most frequent.

• Grouped data: use class marks and frequencies.

• Mean for analysis, median for skewed data.

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Statistics: Mean, Median and Mode to activate prior knowledge.

2. Present the core ideas of Statistics: Mean, Median and Mode with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Statistics: Mean, Median and Mode.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Statistics: Mean, Median and Mode and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Statistics: Mean, Median and Mode for classroom display.

Assessment Questions

1. Find the mean of the given frequency distribution.

2. Find the median and mode of the data.

3. When is the median better than the mean?

Homework

• Write a one-page summary of Statistics: Mean, Median and Mode in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)

Unit 7: Index Numbers

Learning Objectives

• Explain the meaning and uses of index numbers.

• Construct a simple price index.

• Interpret the Consumer Price Index.

Key Concepts

• Index numbers measure relative change over time.

• Base year = 100; simple aggregative method.

• Weighted index: Laspeyres and Paasche formulas.

• CPI measures inflation faced by households.

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Index Numbers to activate prior knowledge.

2. Present the core ideas of Index Numbers with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Index Numbers.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Index Numbers and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Index Numbers for classroom display.

Assessment Questions

1. Construct a price index for the given commodities.

2. What is a base year? Why is it chosen carefully?

3. Interpret: CPI rose from 150 to 165.

Homework

• Write a one-page summary of Index Numbers in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)

Unit 8: Probability: Basic Concepts

Learning Objectives

• Define probability and its range 0 to 1.

• Calculate probability of simple events.

• Apply the addition rule for mutually exclusive events.

Key Concepts

• Probability = favourable outcomes / total outcomes.

• Sample space and events; certain vs impossible.

• Addition rule: P(A or B) = P(A) + P(B).

• Business uses: risk, quality control, insurance.

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Probability: Basic Concepts to activate prior knowledge.

2. Present the core ideas of Probability: Basic Concepts with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Probability: Basic Concepts.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Probability: Basic Concepts and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Probability: Basic Concepts for classroom display.

Assessment Questions

1. Find P(getting a 6) on a fair die.

2. A bag has 3 red and 5 blue balls. Find P(red).

3. Find P(king or queen) from a pack of cards.

Homework

• Write a one-page summary of Probability: Basic Concepts in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)